ORACLE CLOUD

Budgetary Control and Encumbrance Accounting Configuration Best Practices

ORACLE WHITE PAPER | JUNE 2016

ORACLE

Table of Contents

Introduction 3

Budgetary Control vs. Encumbrance Accounting 3

Budgetary Control 4

- Budgetary Control Validation Control Level 4
- Funds Available 5
  - Budget Calendar 5
  - Budget Calendar Example 5
  - Budget Account 7
  - Budget Account Example 8
  - Budget Calendar and Budget Accounts 10
- Budget vs. Consumption Inquiry and Reporting 10

Encumbrance Accounting 11

Budgetary Control and Encumbrance Accounting 12

General Ledger Only 12

Implementation Example 13

- Transaction View 18
- Budget View 19
- Encumbrance Accounting 23

Other Considerations 25

- Manage Budgetary Control Choices 25
- Supplemental Rules 25
- Source Detail and Summary Control Budgets 26

Release Budget for Consumption 27

Reporting Trees 28

Implementing During a Fiscal Year 28

Conclusion 28

Additional Resources 28

Introduction

Oracle ERP Cloud is highly configurable to meet public sector budgetary control and accounting requirements. The General Ledger, Subledger Accounting, Budgetary Control and encumbrance accounting features provide a comprehensive solution to meet a broad spectrum of public sector requirements. This document reviews how these features can be configured, and provides specific sample implementations.

Budgetary Control vs. Encumbrance Accounting

In Oracle ERP Cloud, Budgetary Control and encumbrance accounting are two independent features. Budgetary Control refers to the group of system options and the processes used to determine which transactions are subject to validation against budgets and budget consumption to prevent overspending. Encumbrance accounting is an accounting practice that creates journal entries for requisitions and purchase orders that will become expenditures when goods and services are invoiced or received. The recording of estimated costs before the actual expenditures allows managers to plan for the future impact of previous financial decisions.

You can implement Budgetary Control and encumbrance accounting together or independently depending on your organization's business requirements. The following are the four most common implementation decisions:

- Real-time spending controls - Budgetary Control
- General Ledger journals for requisitions and purchase orders - encumbrance accounting
- Real-time spending controls and General Ledger journals for requisitions and purchase orders - Budgetary Control and encumbrance accounting
- Budget vs. actual reporting using General Ledger - no Budgetary Control or encumbrance accounting

Review the following decision points to determine which features to enable:

BUDGETARY CONTROL VS. ENCUMBRANCE ACCOUNTING DECISION POINTS

Decision Point	Budgetary Control Required?	Encumbrance Accounting Required?
Do you want transaction amounts (from requisitions, purchase orders, invoices, receipts and journals) to be validated against a budget during transaction processing in real time?	Yes	No
Do you want journals in General Ledger to represent purchase order obligations or requisition commitments?	No	Yes
Do you want to report from General Ledger purchase order obligation or requisition commitment balances?	No	Yes
Do you want to report budget vs. actual expenditures only?	No	No

Use only Budgetary Control when there's no requirement to have general ledger journals or balances for purchase orders and/or requisitions, and you need real-time spending controls on transactions.

Use only encumbrance accounting when there's no requirement for validating transaction amounts against the budget during transaction processing, and you need to report purchase orders and/or requisition amounts.

Use Budgetary Control and encumbrance accounting when there's a requirement for real-time spending controls of transactions amounts, general ledger journals, or balances for purchase orders, and/or requisitions.

Use General Ledger when you only need to report the budget vs. actual expenditures periodically. You don't need to implement the Budgetary Control and encumbrance accounting features.

Budgetary Control

Implement Budgetary Control when there's a requirement for real-time spending controls on transaction amounts. Consider the following when implementing Budgetary Control:

- Level of control for budgetary control validation (How tightly are funds controlled?)
- Funds available (How much can I spend?)
- Level of detail for budget vs. consumption inquiry and reporting

Assess your spending control and reporting needs and create a control budget with the budget dimensions and setup options selected for budgetary control validation. Budget and funds available balances are maintained by the budget dimensions you select, such as ledger, calendar, periods, and budget segments. Define setup options such as control level, budget source, and tolerance, in addition to other details.

Budgetary Control Validation Control Level

There are four levels to control spending against a transaction during budgetary control validation. The following table describes the four control levels with their respective actions:

BUDGETARY CONTROL LEVEL

Control Level	Action Taken for Insufficient Funds for a Transaction
Absolute	Stops transaction processing and does not record transaction against the budget. Prevents transaction from continuing until insufficient funds are resolved.
Advisory	Provides warning at transaction processing and records transaction against the budget. Allows transaction to continue.
Track	Records transaction against the budget. Allows transaction to continue.
None	Processes transaction normally but does not record it against the budget. Does not provide a message to the transaction user.

Funds Available

The funds available amount is calculated as the budget amount minus the budget consumption amount (or the funds reserved amount) for a given control budget, budget period, and budget account combination.

Use the following attributes to determine the funds available amount when budgetary control validation is performed on a transaction or journal:

- Budget date determines the budget period
- Distribution account determines the budget account (for example, charge account for a requisition and purchase order)
- Ledger and budget period determine the control budget

Funds available = budget balance – consumption within one budget account for one budget period

Budget Calendar

The budget calendar assigned to the control budget determines how much you can spend in each period. You can either share calendars between General Ledger and Budgetary Control, or designate a calendar for Budgetary Control only. Consider how you define your budget calendar since the period interval will affect your funds available as budget amounts are summarized according to the calendar period interval.

Use a yearly calendar and/or a monthly calendar to achieve the desired budget available for consumption. The following example shows how the calendar period interval affects the funds available.

Budget Calendar Example

Suppose you have defined two calendars: one with a yearly period interval and another with a monthly period interval:

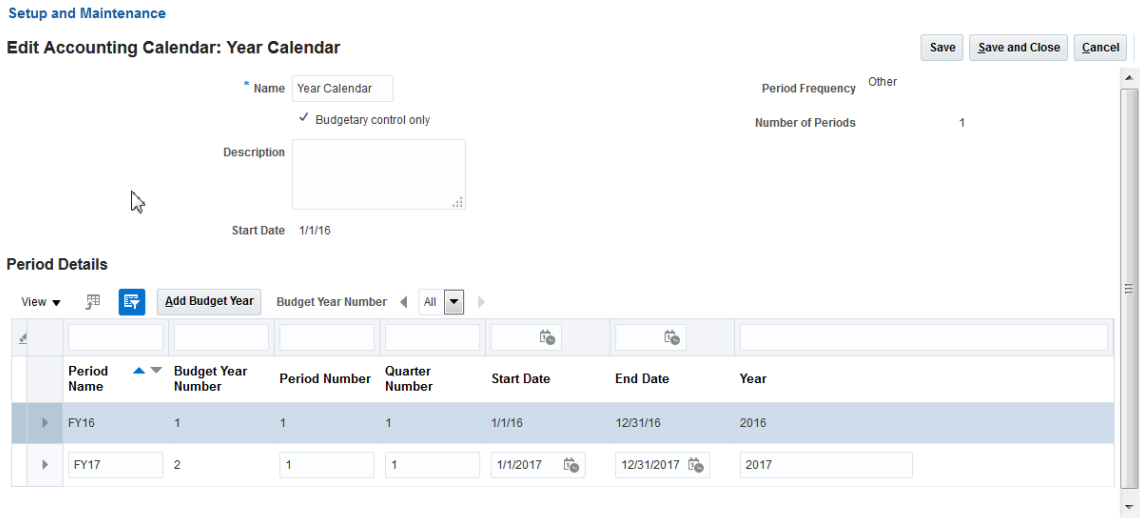
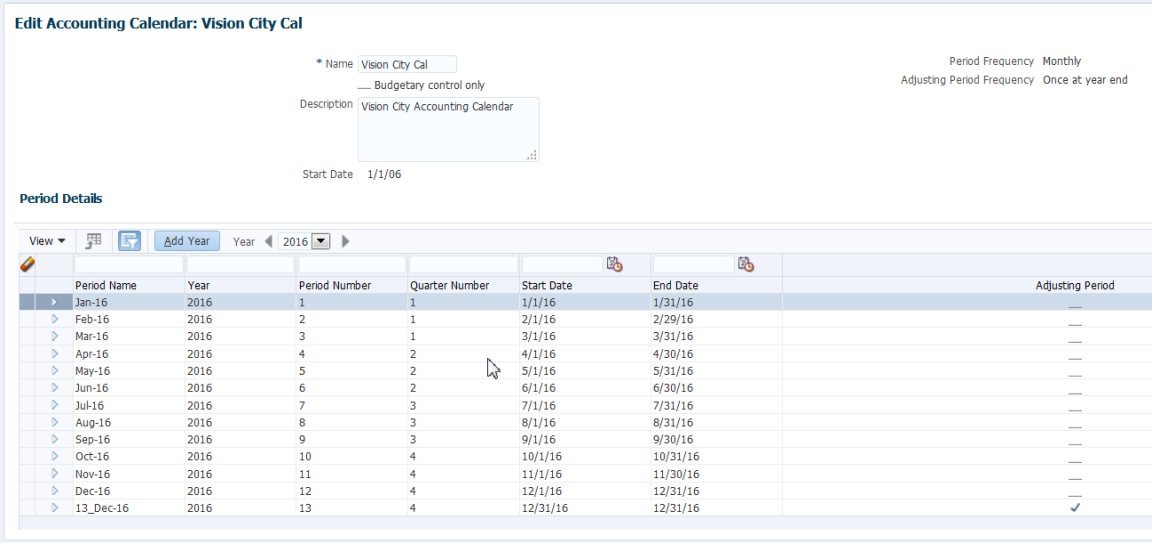


Figure 1. Yearly Calendar



Edit Accounting Calendar: Vision City Cal

* Name Vision City Cal
Budgetary control only
Description Vision City Accounting Calendar
Start Date 1/1/06

Period Frequency Monthly
Adjusting Period Frequency Once at year end

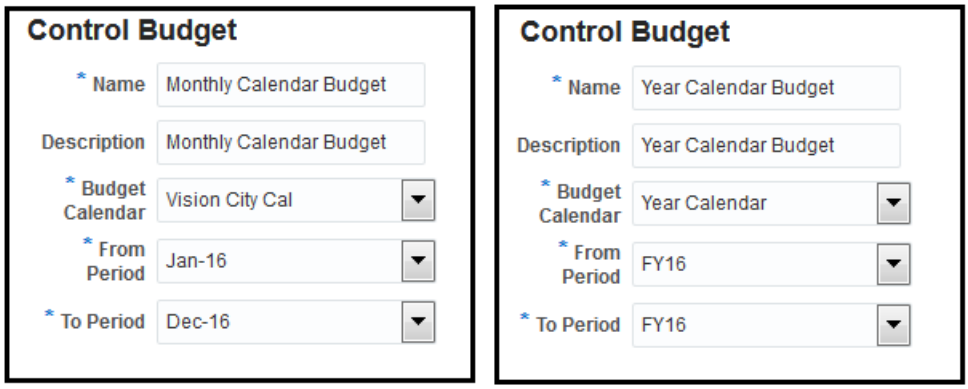
Period Details

View Add Year Year 2016

Period Name	Year	Period Number	Quarter Number	Start Date	End Date	Adjusting Period
Jan-16	2016	1	1	1/1/16	1/31/16	
Feb-16	2016	2	1	2/1/16	2/29/16	
Mar-16	2016	3	1	3/1/16	3/31/16	
Apr-16	2016	4	2	4/1/16	4/30/16	
May-16	2016	5	2	5/1/16	5/31/16	
Jun-16	2016	6	2	6/1/16	6/30/16	
Jul-16	2016	7	3	7/1/16	7/31/16	
Aug-16	2016	8	3	8/1/16	8/31/16	
Sep-16	2016	9	3	9/1/16	9/30/16	
Oct-16	2016	10	4	10/1/16	10/31/16	
Nov-16	2016	11	4	11/1/16	11/30/16	
Dec-16	2016	12	4	12/1/16	12/31/16	
13_Dec-16	2016	13	4	12/31/16	12/31/16	✓

Figure 2. Monthly Calendar

Two control budgets are created and each is assigned with a different budget calendar. The Monthly Calendar Budget uses the monthly calendar and the Year Calendar Budget uses the yearly calendar.



Control Budget

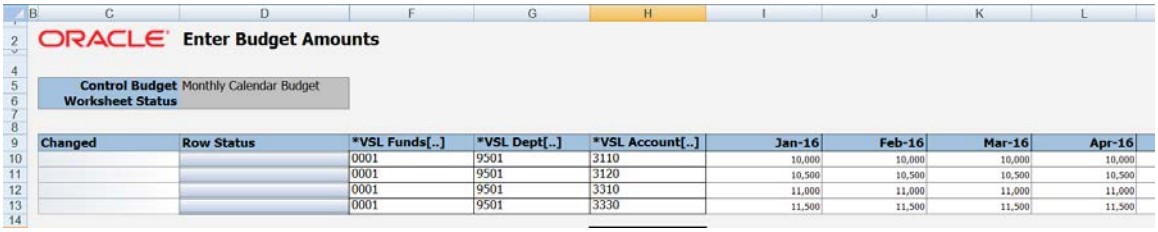
* Name Monthly Calendar Budget
Description Monthly Calendar Budget
* Budget Calendar Vision City Cal
* From Period Jan-16
* To Period Dec-16

Control Budget

* Name Year Calendar Budget
Description Year Calendar Budget
* Budget Calendar Year Calendar
* From Period FY16
* To Period FY16

Figure 3. Control Budgets: Monthly Calendar Budget and Year Calendar Budget

Budget amounts are imported into the Monthly Calendar Budget. Since the Monthly Calendar Budget and the Year Calendar Budget are linked, budget amounts are loaded into both budgets at import time. For more information on linking control budgets, see the Source Detail and Summary Control Budgets section.



ORACLE Enter Budget Amounts

Control Budget Worksheet Status Monthly Calendar Budget

Changed	Row Status	*VSL Funds[..]	*VSL Dept[..]	*VSL Account[..]	Jan-16	Feb-16	Mar-16	Apr-16
		0001	9501	3110	10,000	10,000	10,000	10,000
		0001	9501	3120	10,500	10,500	10,500	10,500
		0001	9501	3310	11,000	11,000	11,000	11,000
		0001	9501	3330	11,500	11,500	11,500	11,500

Figure 4. Monthly Calendar Budget Import Amounts

In the following figure, the funds available results are summarized using the yearly interval in the Year Calendar Budget, and remain at the monthly interval in the Monthly Calendar Budget.

Ledger Vision City **Control Budget** Monthly Calendar Budget **Currency** USD

Budget Period: Apr-16, Aug-16, Dec-16, Feb-16, Jan-16, Jul-16

Fund: 0001

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Jan-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Feb-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Mar-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Apr-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	May-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Jun-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Jul-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Aug-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Sep-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Oct-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Nov-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Dec-16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00 %

Ledger Vision City **Control Budget** Year Calendar Budget **Currency** USD

Budget Period: FY16

Fund: 0001

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	FY16	120,000.00	120,000.00	0.00	0.00	0.00	0.00	0.00	0.00	120,000.00	100.00%
0001-9501-3120	Major Fund-XCC PM Department 9501-Vehicle Usage	FY16	126,000.00	126,000.00	0.00	0.00	0.00	0.00	0.00	0.00	126,000.00	100.00%
0001-9501-3310	Major Fund-XCC PM Department 9501-Uniforms & Safety Equipme	FY16	132,000.00	132,000.00	0.00	0.00	0.00	0.00	0.00	0.00	132,000.00	100.00%
0001-9501-3330	Major Fund-XCC PM Department 9501-Professional Development	FY16	138,000.00	138,000.00	0.00	0.00	0.00	0.00	0.00	0.00	138,000.00	100.00%

Figure 5. Monthly Calendar Budget and Year Calendar Budget Funds Available Results

The budget account 0001-9501-3110 has $10,000 loaded per month. The funds available for the Monthly Calendar Budget is $10,000 per month, while the Year Calendar Budget is $120,000 for the entire year.

Budget Account

The budget account consists of budget segments assigned to the control budget structure that represents the spending controls and/or report spending dimensions. These can be general ledger accounting segments. You can share the chart of accounts and their segment value sets between General Ledger and Budgetary Control. You can select a subset of segments you want the budget dimension to represent.

The control budget structure also defines the level of summarization at which you control or report spending. Use tree hierarchies with tree labels to assign the level of summarization in the budget segment.

- If you don't assign tree hierarchies, the budget amounts are stored at the detail level.
- If you assign a tree hierarchy and label to a budget segment, the budget amounts are stored and summarized at the defined level.

The budgetary control process maps the distribution's account (for example, the charge account on a requisition or purchase order) and budget date to the control budget's budget account and budget period, respectively, to determine the funds available. Consider the assignment of budget segments and associated tree hierarchies and labels to achieve the correct funds available amount, and enforce spending controls and reporting.

Budget Account Example

The following example shows how trees can affect the funds available amount. In the scenario, Vision City ledger has a chart of accounts with the following segments:

Fund – Department - Program – Account – Project – Location

Budget and spending controls are for the fund – department – account segments. The account segment has a tree hierarchy defined in the following figure. Tree labels have been assigned to parent accounts at different levels within the tree to achieve the summarization needed for spending controls and reporting.

Manage Trees and Tree Versions

Tree Structure Code GL_ACCT_FLEX

Tree Code VSL Account

Tree Version Name V1

Actions ▾ View ▾ Format ▾ Freeze Detach Wrap

Node Name	Node Description	Label
1000	Total Expenses	Account Level 3
0100	General Expenditures	
1100	Operating Expense	Account Level 2
1800	Total Cost of Goods Sold	Account Level 1
2000	Personnell	Account Level 1
3100	Office Expenses	Account Level 1
3110	Insurance	
3120	Vehicle Usage	
3130	Equipment Maintenance	
3140	Utilities	
3300	Other Employee Expenses	Account Level 1
3310	Uniforms & Safety Equipment	
3320	Professional Services	
3330	Professional Development	

Figure 6. Account Tree Hierarchy

Use this tree hierarchy to create control budgets to achieve various funds available amounts. Two control budgets have been created.

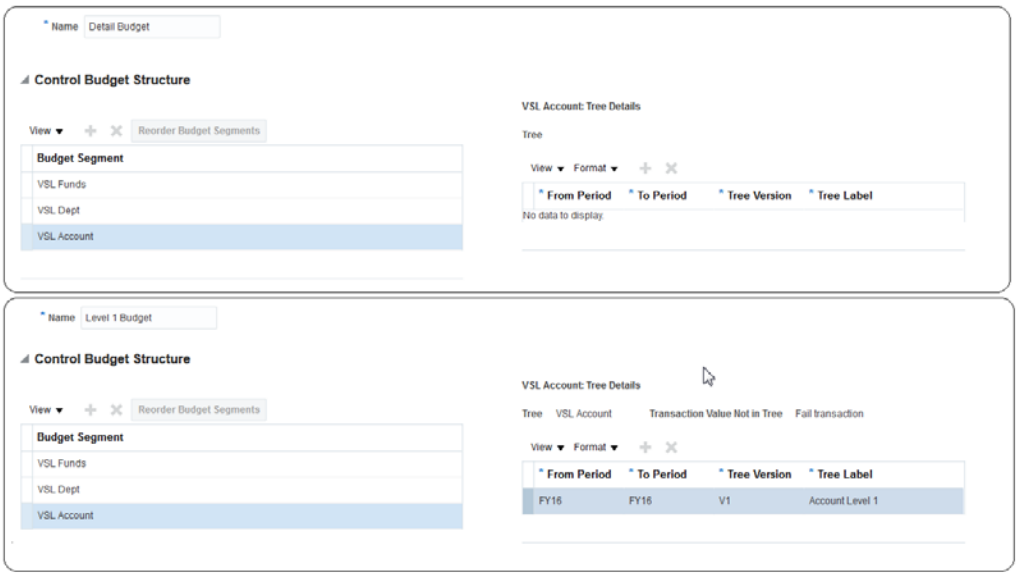
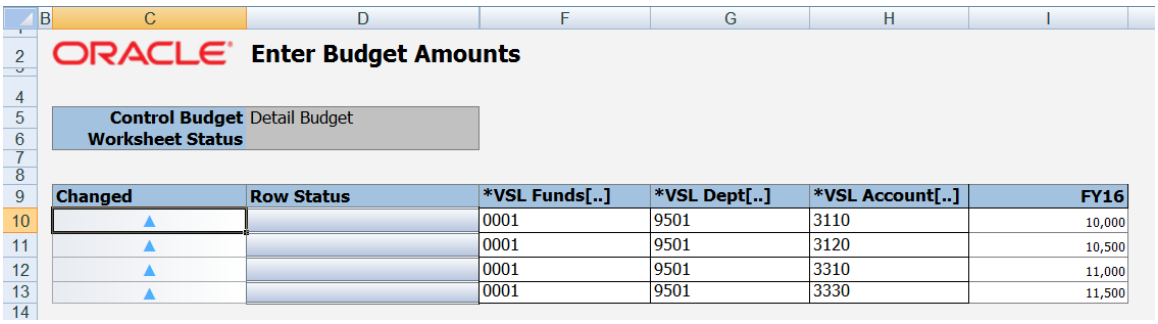


Figure 7. Control Budgets: Detail Budget and Level 1 Budget

The Detail Budget doesn't have a tree assigned to the account segment, so balances are stored at the detail level. The Level 1 Budget is assigned the Account tree and uses the Account Level 1 label; therefore balances are stored by the type of operating expense.

Budget amounts are loaded into the Detail Budget at the detail level. Since the Detail Budget and the Level 1 Budget are linked, budget amounts are loaded into both budgets at import time. For more information on linking control budgets, see the Source Detail and Summary Control Budgets section.



Changed	Row Status	*VSL Funds[..]	*VSL Dept[..]	*VSL Account[..]	FY16
▲		0001	9501	3110	10,000
▲		0001	9501	3120	10,500
▲		0001	9501	3310	11,000
▲		0001	9501	3330	11,500

Figure 8. Detail Budget Import Amounts

The Detail Budget amounts are at the detail account level (for example, accounts 3110, 3120, 3310, and 3330). The Level 1 Budget amounts are summarized to the tree label Account Level 1, so these amounts are reflected in the budget amounts for office expenses (3000) and other employee expenses (3300).

Ledger Vision City **Control Budget** Detail Budget **Currency** USD

Budget Period: FY16

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	FY16	10,000.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	100.00%
0001-9501-3120	Major Fund-XCC PM Department 9501-Vehicle Usage	FY16	10,500.00	10,500.00	0.00	0.00	0.00	0.00	0.00	0.00	10,500.00	100.00%
0001-9501-3310	Major Fund-XCC PM Department 9501-Uniforms & Safety Equipme	FY16	11,000.00	11,000.00	0.00	0.00	0.00	0.00	0.00	0.00	11,000.00	100.00%
0001-9501-3330	Major Fund-XCC PM Department 9501-Professional Development	FY16	11,500.00	11,500.00	0.00	0.00	0.00	0.00	0.00	0.00	11,500.00	100.00%

Ledger Vision City **Control Budget** Level 1 Budget **Currency** USD

Budget Period: FY16

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3100	Major Fund-XCC PM Department 9501-Office Expenses	FY16	20,500.00	20,500.00	0.00	0.00	0.00	0.00	0.00	0.00	20,500.00	100.00%
0001-9501-3300	Major Fund-XCC PM Department 9501-Other Employee Expenses	FY16	22,500.00	22,500.00	0.00	0.00	0.00	0.00	0.00	0.00	22,500.00	100.00%

Figure 9. Budget Amounts in the Detail Budget and Level 1 Budget

Budget Calendar and Budget Accounts

A control budget structure and the period interval of the calendar determine the funds available. The combination of the budget calendar period interval, the budget segments and the use of tree hierarchy and labels determines the budget accounts and amount of funds available for a control budget.

Budget vs. Consumption Inquiry and Reporting

You may need multiple control budgets to meet your controlling funds and reporting needs. Start with examples of your statutory and management reports and work backwards to determine the budget segments, hierarchies and budget time intervals needed in constructing control budgets for reporting purposes. Budgetary control (that is, how much budget is available for consumption per budget account and budget period) doesn't need to use the same budget accounts and periods for management or statutory reporting.

Since balances are maintained based on the budget periods and tree hierarchy labels assigned to the control budget, reporting and inquiry is available only at that level or above. For example, if a control budget is set to control at a certain label, you can report on any budget accounts summarized at a higher level. However, balances at a lower level won't be available. Similarly, if a control budget has a monthly calendar, the system will summarize monthly balances into quarterly and yearly balances for reporting. However, if a control budget has an annual calendar, then monthly and quarterly balances aren't available for that control budget.

You can control funds at a detailed budget account level and report or inquire on budget accounts at a summary level by adding tree hierarchies to an Essbase cube solely for reporting. For more information on reporting trees, see the Reporting Trees section.

To report or inquire on more detailed accounts and periods than you are required to control funds, create control budgets with a control level of **Track**.

*Note: Control budgets with a control level of **Track** record transactions against a budget but doesn't consider funds available.*

Encumbrance Accounting

Use the encumbrance accounting feature to:

- Report on General Ledger commitment and obligation balances for requisitions and purchase orders
- Allow transactions with insufficient funds to continue on the transaction life cycle and be approved, validated or posted
- Allow transaction users to continue without being notified of insufficient funds
- Provide budget vs. consumption reporting that coincides with the accounting dimensions

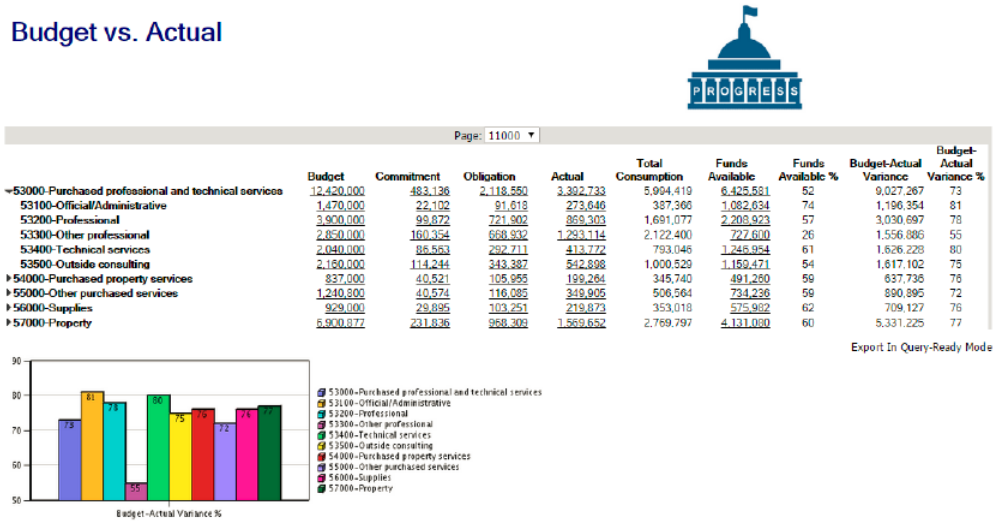
To use encumbrance accounting, you must enable Budgetary Control and encumbrance accounting for the ledger and business units, and assign the accrual with encumbrance accounting and subledger accounting method to your ledger. Setting up control budgets isn't necessary.

Since Budgetary Control isn't used for reporting on budget vs. actual, budget amounts must be loaded and maintained in the General Ledger balances Essbase cube. Encumbrance accounting doesn't track and store the funds available balance. Encumbrance balances for commitments, obligations, and actual expenditures are stored in the General Ledger balances Essbase cube to be used for analysis and financial reports. You can calculate the funds available in a report. Use tree hierarchies in the General Ledger balances Essbase cube for summary reporting.

Budget balances are loaded using one of the following methods:

- Create Budgets in Spreadsheet option in the General Accounting Dashboard
- Import General Ledger budget balances feature using the GeneralLedgerBudgetBalanceImportTemplate.xlsm template

In General Ledger, you can view budget, actual, and encumbrance balances in the General Ledger balances Essbase cube from the Account Monitor, Smart View, or the Financial Reporting Studio/Financial Reporting Center. You can create reports such as the budget vs. actual report shown in the following figure.



	Budget	Commitment	Obligation	Actual	Total Consumption	Funds Available	Funds Available %	Budget-Actual Variance	Budget-Actual Variance %
53000-Purchased professional and technical services	12,420,000	483,136	2,118,550	3,392,733	5,994,419	6,425,581	52	9,027,267	73
53100-Official/Administrative	1,470,000	22,102	91,818	273,646	387,366	1,082,634	74	1,196,354	81
53200-Professional	3,900,000	99,872	721,902	869,303	1,691,077	2,208,923	57	3,030,697	78
53300-Other professional	2,850,000	160,354	668,932	1,293,114	2,122,400	727,600	26	1,556,886	55
53400-Technical services	2,040,000	86,563	292,711	413,772	793,046	1,246,954	61	1,626,228	80
53500-Outside consulting	2,160,000	114,244	343,387	542,898	1,000,529	1,159,471	54	1,617,102	75
54000-Purchased property services	837,000	40,521	105,955	199,264	345,740	491,260	59	637,736	76
55000-Other purchased services	1,240,800	40,574	116,085	349,905	506,564	734,236	59	890,895	72
56000-Supplies	929,000	29,895	103,251	219,873	353,018	575,982	62	709,127	76
57000-Property	6,900,877	231,836	968,309	1,569,652	2,769,797	4,131,080	60	5,331,225	77

Figure 10. Budget vs. Actual Reporting with Encumbrance Accounting

Budgetary Control and Encumbrance Accounting

Enable both Budgetary Control and encumbrance accounting when you need to enforce spending controls and report commitment or obligations from the General Ledger.

General Ledger Only

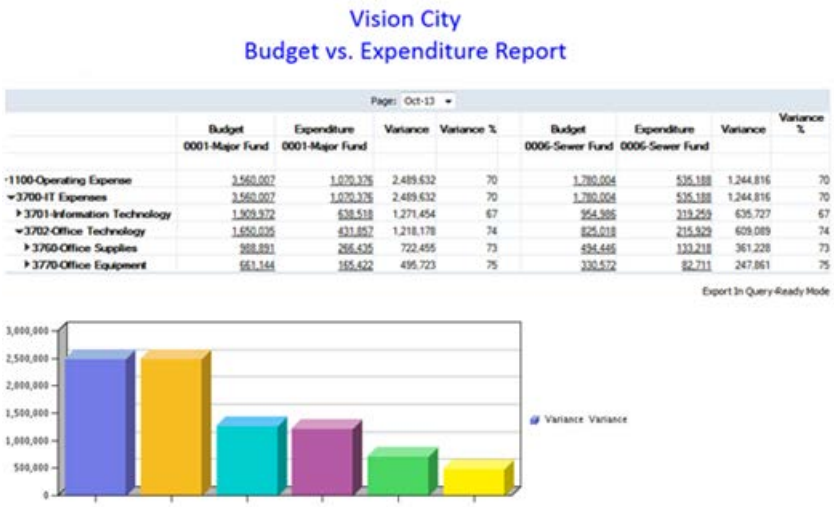
Use General Ledger instead of Budgetary Control and encumbrance accounting if your organization:

- Doesn't need to report on commitment and/or obligation balances
- Doesn't need to enforce spending controls
- Needs only to report budget vs. actual

In this configuration, the standard accrual subledger accounting method must be assigned to your ledger. Budgetary Control and encumbrance accounting are not enabled. Load the budget balances into the General Ledger balances Essbase cube using one of the following methods:

- Create Budgets in Spreadsheet option in the General Accounting Dashboard
- Import General Ledger budget balances feature using the GeneralLedgerBudgetBalanceImportTemplate.xlsm template

Create reports using the budget and actual balances from the General Ledger balances Essbase cube. The following figure displays a sample budget vs. expenditure report:



Vision City

Budget vs. Expenditure Report

Page: Oct-13

	Budget 0001-Major Fund	Expenditure 0001-Major Fund	Variance	Variance %	Budget 0006-Sewer Fund	Expenditure 0006-Sewer Fund	Variance	Variance %
1100-Operating Expense	3,560,007	1,070,375	2,489,632	70	1,780,004	535,188	1,244,816	70
3700-IT Expenses	3,560,007	1,070,375	2,489,632	70	1,780,004	535,188	1,244,816	70
3701-Information Technology	1,909,972	638,518	1,271,454	67	954,986	319,259	635,727	67
3702-Office Technology	1,650,035	431,857	1,218,178	74	825,018	215,929	609,089	74
3760-Office Supplies	988,891	266,435	722,455	73	494,446	133,218	361,228	73
3770-Office Equipment	661,144	165,422	495,723	75	330,572	82,711	247,861	75

Export In Query-Ready Mode

Figure 11. Budget vs. Expenditure Report for General Ledger Only Implementations

Implementation Example

The following implementation example demonstrates a common use case when creating control budgets to meet both reporting and spending controls requirements.

The organization has the following requirements:

- Certain operating expenses (such as office expenses and other employee expenses) can exceed the budget, but the total operating expenses can't go over budget.
- Real-time spending controls are required.
  - Budgeted amount for these operating expenses is enforced.
    - Stops transaction processing when there are insufficient funds available for these operating expenses.
  - Certain operating expenses have their own budget, and use warnings for overspending.
    - Triggers a warning when transaction spending exceeds the funds available for the type of operating expense (such as office expenses and other employee expenses)
- Budget is allocated for the year, but split out monthly. However, the full budget amount for the year can be spent in any month in the year.
- Budget Manager has specific reporting requirements:
  - Detail account reporting by month to review detailed budget consumption
  - Summary account reporting by month on budget vs. consumption to review planned spending and compare against prior periods
  - Summary account reporting by year to review funds available and compare against prior years
- Finance Manager has specific reporting requirements:
  - Financial reports from accounting balances must include commitment and obligations from requisitions and purchase orders.

The organization implements both Budgetary Control and the encumbrance accounting since they need real-time spending controls and encumbrance accounting journals on requisitions and purchase orders.

Terry and Chris are budget managers in the organization. Terry manages the operating expenses budget, and Chris manages the office expenses budget. To simplify the example and focus on the use of tree hierarchy and labels in segments, the organization uses a single fund, 0001, and a single department, 9501. A tree hierarchy with labels, VSL Account, is created with the following account values.

Node Name	Node Description	Label
1000	Total Expenses	Account Level 3
0100	General Expenditures	
1100	Operating Expense	Account Level 2
1800	Total Cost of Goods Sold	Account Level 1
2000	Personnell	Account Level 1
3100	Office Expenses	Account Level 1
3110	Insurance	
3120	Vehicle Usage	
3300	Other Employee Expenses	Account Level 1
3310	Uniforms & Safety Equipment	
3320	Professional Services	

Figure 12. VSL Account Tree Hierarchy

To institute the various spending controls needed and meet their reporting requirements, the organization uses a combination of budget calendar period intervals, for example, monthly period and yearly period, and a tree hierarchy and labels.

Three control budgets are created that all use the fund – department – account segments.

- Monthly Track control budget is defined at a detail level to track monthly consumption. The control budget's balances are used in reporting from detail to summary levels.
- The other two control budgets both have an annual budget period. A tree hierarchy is assigned and tree labels are used to control spending at different levels within the organization of the account segment.
  - Annual Absolute Account Level 2 blocks the approval of transactions when the budget exceeds operating expenses (1100), excluding cost of goods sold and personnel expenses. Terry manages this budget. For more information on how to exclude accounts, see the Other Considerations section.
  - Annual Advisory Account Level 1 provides a warning when overspending occurs based on the type of operating expense (3100 for office expenses, 3300 for other employee expenses). Chris manages the office expense portion of this budget.

The following describes the setup of the control budgets:

Control Budget Setup		
Monthly Track	Annual Absolute Account Level 2	Annual Advisory Account Level 1
» Tracks spending and loading of budget amounts » No trees are used so amounts are stored at detail level » 12 periods per year	» Blocks overspending » Tree label is account level 2, therefore spending controls are at the operating expenses level » One period per year	» Warns if overspending occurs » Tree label is account level 1, therefore spending controls are based on the type of operating expenses level (such as office expenses, or other employee expenses) » One period per year

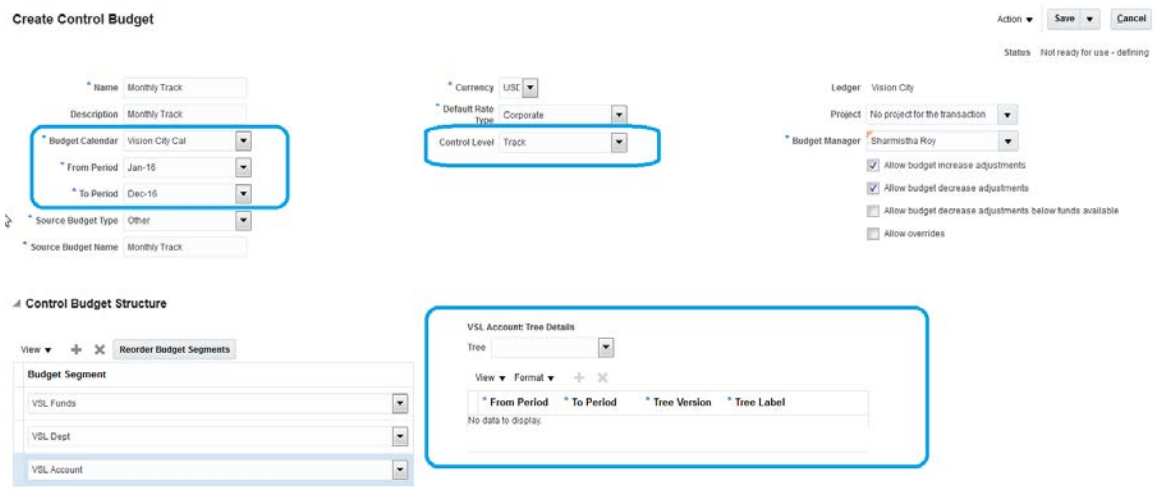


Figure 13. Monthly Track Control Budget – No Tree Details Assigned

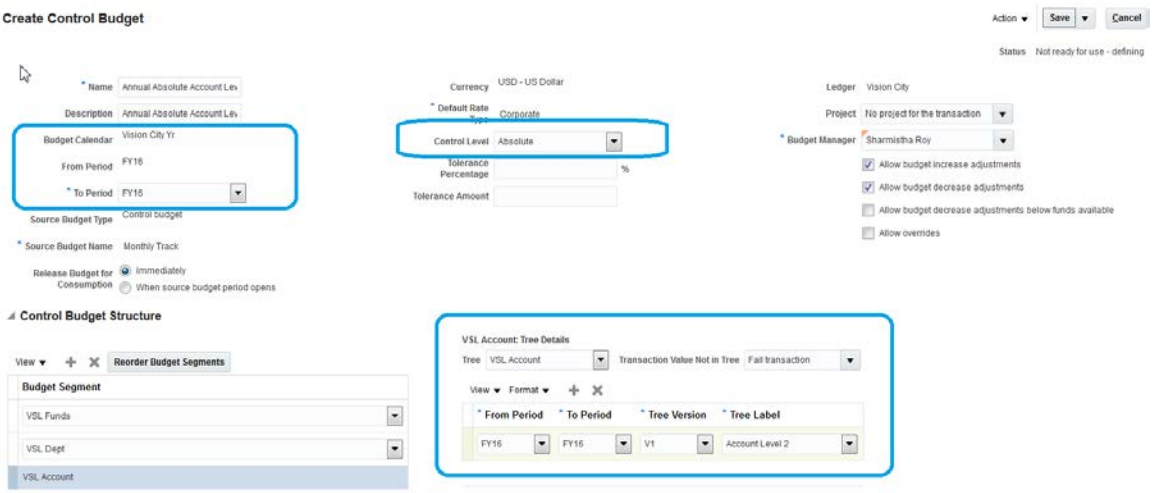


Figure 14. Annual Absolute Account Level 2 – Tree Details Assigned at Level 2 – Operating Expenses Level

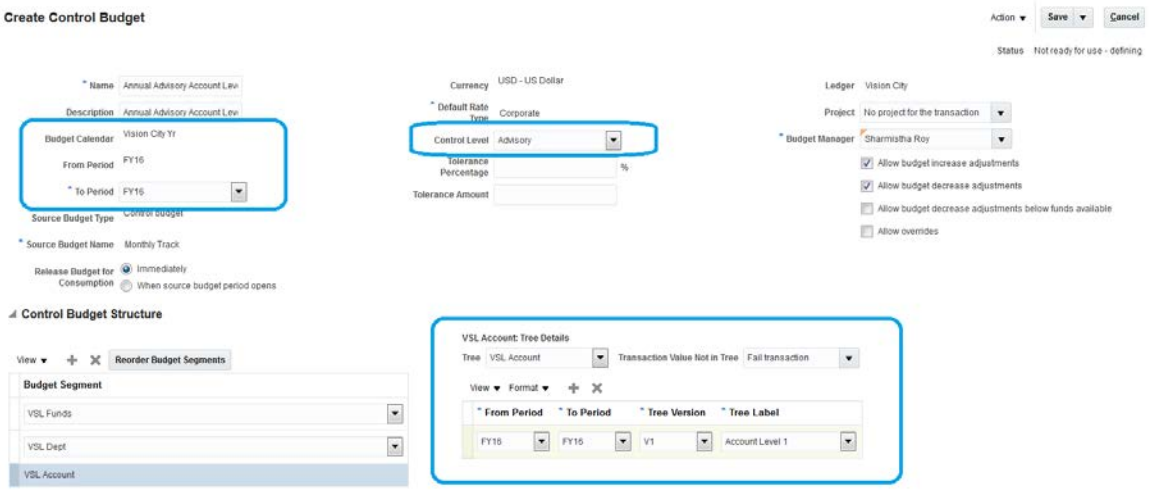
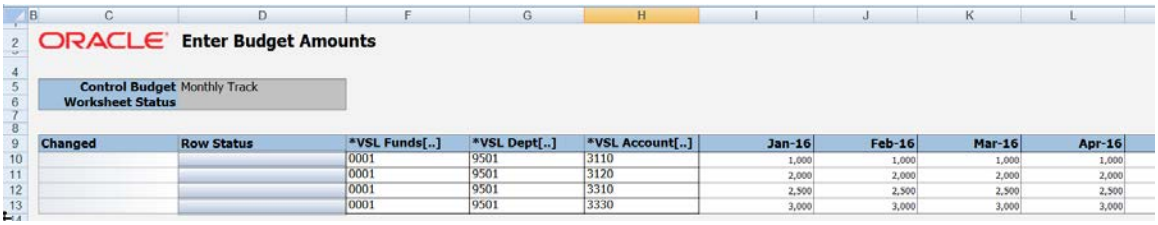


Figure 15. Annual Advisory Account Level 1 – Tree Details Assigned at Level 1 – Type of Operating Expense

The yearly budgets are allocated to budget amounts for each month and are imported into the Monthly Track budget. Since the two yearly budgets are linked to the Monthly Track budget, importing to this budget also loads budget amounts to the two yearly budgets.



ORACLE Enter Budget Amounts

Control Budget Monthly Track
Worksheet Status

Changed	Row Status	*VSL Funds[..]	*VSL Dept[..]	*VSL Account[..]	Jan-16	Feb-16	Mar-16	Apr-16
		0001	9501	3110	1,000	1,000	1,000	1,000
		0001	9501	3120	2,000	2,000	2,000	2,000
		0001	9501	3310	2,500	2,500	2,500	2,500
		0001	9501	3330	3,000	3,000	3,000	3,000

Figure 16. Budget Import of Monthly Track

Budget amounts are imported into the Monthly Track and both annual budgets.

- Annual Advisory Account Level 1 amounts are summarized to the label Account Level 1 for the year so budget amounts are allocated for office expenses (3100) and other employee expenses (3300).
- Annual Absolute Account Level 2 amounts are summarized to the label Account Level 2 and for the year so budget amounts are allocated for the total operating expense level (1100).
- Monthly Track budget remains at the detail monthly amount.

The budget import results are shown below.

Ledger Vision City **Control Budget** Monthly Track **Currency** USD

Budget Period: Jan-16, Apr-16, Aug-16, Dec-16, Feb-16, Jul-16

Fund: 0001

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3110	Major Fund-XCC PM Department 9501-Insurance	Jan-16	1,000.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	1,000.00	100.00%
0001-9501-3120	Major Fund-XCC PM Department 9501-Vehicle Usage	Jan-16	2,000.00	2,000.00	0.00	0.00	0.00	0.00	0.00	0.00	2,000.00	100.00%
0001-9501-3310	Major Fund-XCC PM Department 9501-Uniforms & Safety Equipme	Jan-16	2,500.00	2,500.00	0.00	0.00	0.00	0.00	0.00	0.00	2,500.00	100.00%
0001-9501-3330	Major Fund-XCC PM Department 9501-Professional Development	Jan-16	3,000.00	3,000.00	0.00	0.00	0.00	0.00	0.00	0.00	3,000.00	100.00%

Ledger Vision City **Control Budget** Annual Advisory Account Level 1 **Currency** USD

Budget Period: FY16

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-3100	Major Fund-XCC PM Department 9501-Office Expenses	FY16	36,000.00	36,000.00	0.00	0.00	0.00	0.00	0.00	0.00	36,000.00	100.00%
0001-9501-3300	Major Fund-XCC PM Department 9501-Other Employee Expenses	FY16	66,000.00	66,000.00	0.00	0.00	0.00	0.00	0.00	0.00	66,000.00	100.00%

Ledger Vision City **Control Budget** Annual Absolute Account Level 2 **Currency** USD

Budget Period: FY16

Funds Available by Budget Account

Budget Account	Budget Account Description	Budget Period	Budget Balance	Budget Initial Balance	Budget Adjustment Balance	Total Consumption Balance	Commitment Balance	Obligation Balance	Expenditure Balance	Other Encumbered Balance	Funds Available Balance	Funds Available Percent
0001-9501-1100	Major Fund-XCC PM Department 9501-Operating Expense	FY16	102,000.00	102,000.00	0.00	0.00	0.00	0.00	0.00	0.00	102,000.00	100.00%

Figure 17. Budget Import Results

Using the budget accounts highlighted in the above import results table, the funds available for each budget are described below. The various funds available balances are achieved through the use of different period intervals and tree hierarchies and labels.

Control Budget	Budget Account	Budget Period	Funds Available
Monthly Track	0001-9501-3110	Jan-2016 – Dec-2016	$1,000 in each period
Annual Advisory Account Level 1	0001-9501-3100	FY16	$36,000 Includes amounts for accounts 3110 and 3130
Annual Absolute Account Level 2	0001-9501-1100	FY16	$102,000 Includes amounts for accounts 3100 and 3300

After the import is complete, the control budget status is changed to **In Use** so the budget is available for consumption. Transactions are entered in the subledgers and journals in General Ledger. The funds available are consumed when the budgetary control reservation validation is successful.

For example, a payables clerk enters an invoice for insurance using account 3110. After the clerk performs a funds check by selecting **Invoice Actions** > **Check funds**, the budgetary control validation process uses the following attributes of the transaction to find an applicable control budget and budget account to determine the funds available:

- Distribution account (to determine budget account)
- Budget date (to determine budget period)
- Business unit (to determine if Budgetary Control is enabled and identify the ledger that determines the control budget)

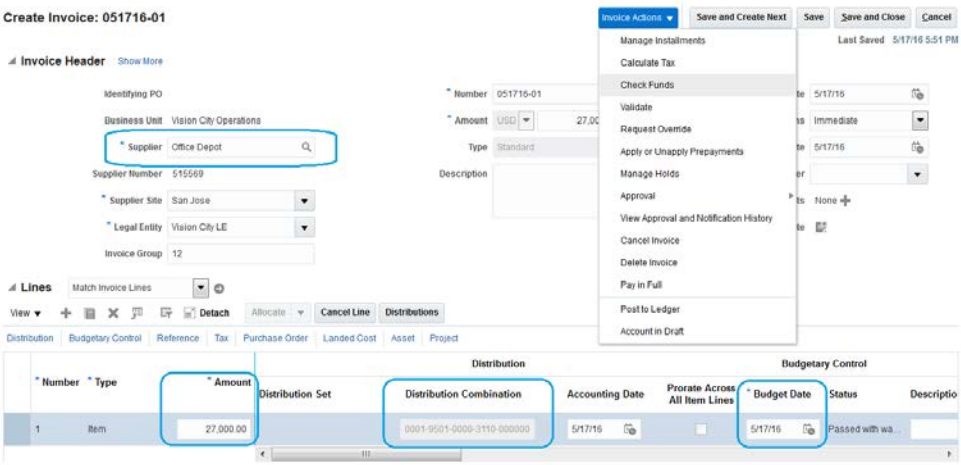


Figure 18. Funds Checking a Payables Invoice

When the funds check process is complete, the budgetary control results can be reviewed using the following two views:

- Transaction view
- Budget view

Transaction View

The transaction view of the funds check shows the funds available impact if the invoice is reserved. This view displays the control budget, budget period, and budget account against which the funds would be reserved. The payables clerk is informed that the invoice distribution has insufficient funds for the office expense (3100) in FY16. The Annual Advisory Account Level 1 funds available are over budget and the application displays an advisory warning of insufficient funds status. Operating expense budget (1100) has sufficient funds for FY16 for the Annual Absolute Account Level 2. Insurance (3110) expense is reserved in May against Monthly Track budget.

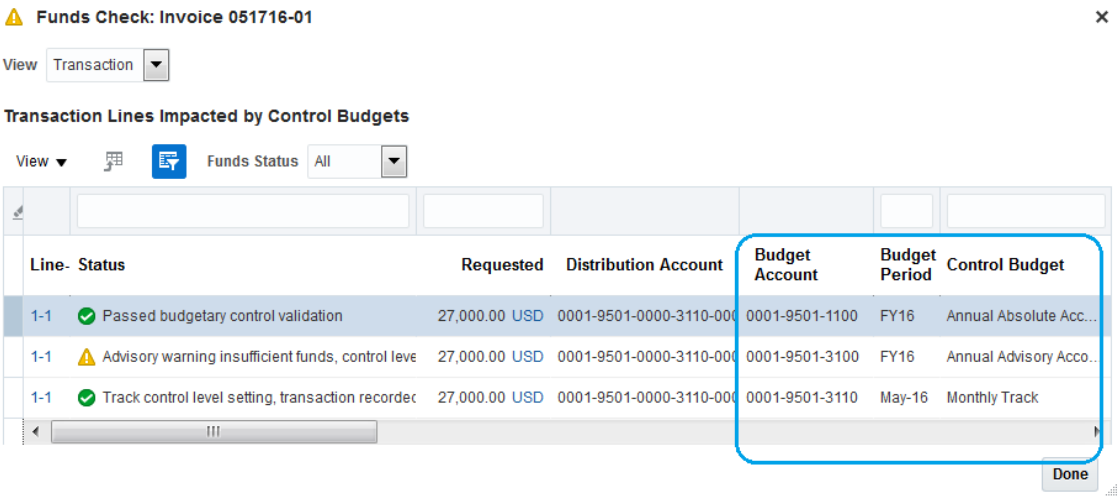


Figure 19. View Results: Transaction View Funds Check

Budget View

The budget view shows the current funds available amounts of the impacted budgets. Using this view, the payables clerk can tell that the Annual Advisory Account Level 1 budget only has $25,205 funds available while the transaction requested amount is $27,000. The budget view also shows that the Monthly Track budget will exceed the funds available for insurance (3110) since there is only $800 of funds available. However the overall operating budget (1100) has $91,205 funds available, which will be sufficient to cover the requested $27,000.

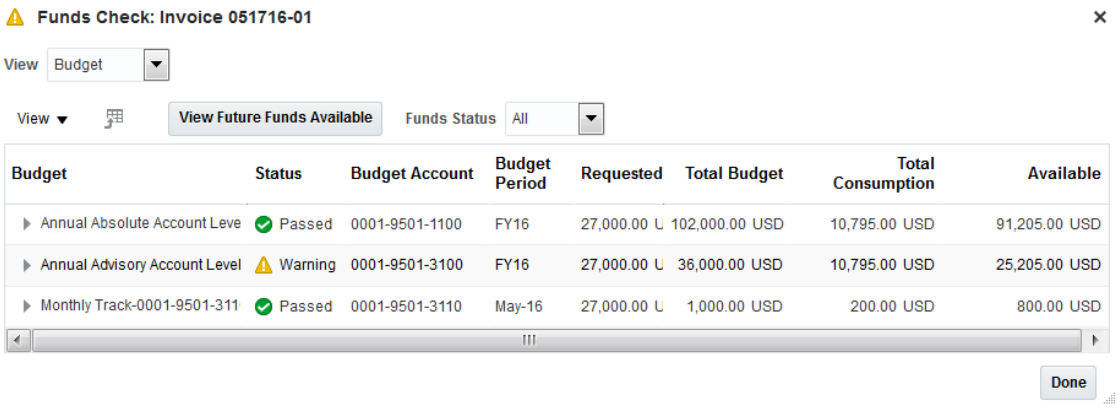


Figure 20. View Results: Budget View Funds Check

The payables clerk validates the invoice, and the funds are reserved against the three budgets. The funds available are decreased by $27,000.00 on each budget.

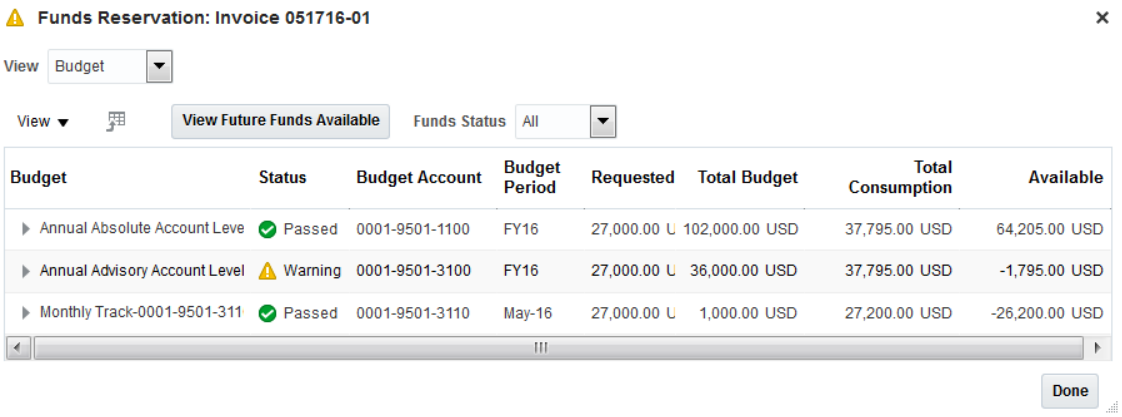


Figure 21. View Results: Budget View Reservation

Budget managers have several configurable tools to keep abreast of the funds status of their respective budgets. These tools are:

- Budget Manager dashboard
- Budgetary Control infolets
- Ad hoc analysis
  - Budgetary Control Real-Time Transactions OTBI
  - Smart View
- Standard reports
  - Budgetary Control Analysis
  - Budgetary Control Exceptions
- Financial reporting

Terry and Chris decide to use the Budgetary Control infolets as they provide quick visuals representing the state of their accounts on the home page. They can also easily drill down from the high level view into the Budget Manager dashboard.

Budgetary Control infolets can be customized for budget managers to monitor funds availability and funds consumption of their critical budgets and budget accounts. Budget managers can use the infolets to gain quicker access to the summarized funds information and further investigate the funds availability in more detail.

The Budget Consumed infolet displays the budget consumption for a predefined group of budget accounts. You can expand the infolet to see the budget accounts with the highest consumption percentages for the control budget. In the expanded view, you can also view the funds available details by clicking the budget account link. Drill down to the Budget Monitor to see more details of the funds available.

Terry uses the Budget Consumed infolet on his home page to monitor the funds available for the operating expenses budget. On Terry's home page, he sees that the operating expense budget is fine, but an alert informs him that there are some accounts that are over budget. The Annual Advisory Account Level 1 budget (the operating expense budget) has $64,200 as funds available. After expanding the infolet, Terry sees that the office expense budget account is $1,800 over budget to date. He contacts Chris, the office expense budget owner, to investigate.

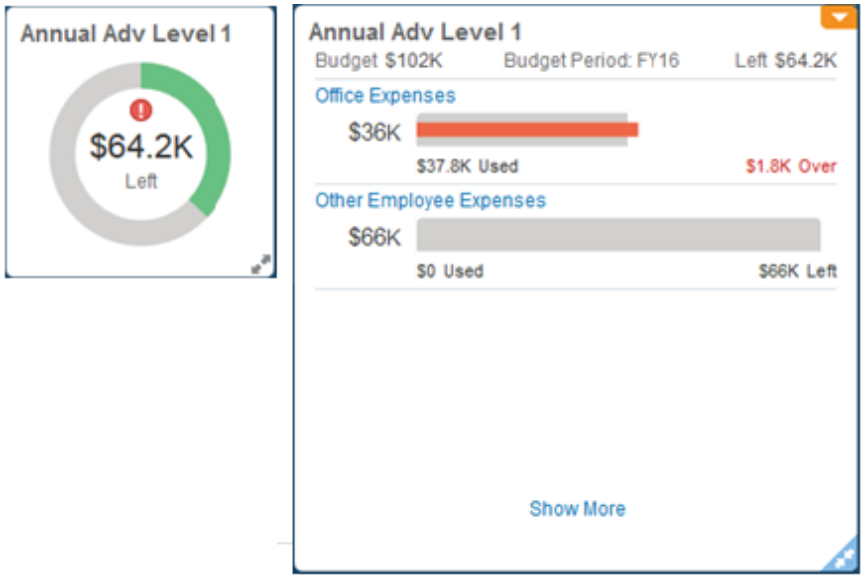


Figure 22. Terry's Home Page - Budget Consumed Infolet

Chris uses the Funds Available infolet on his home page to monitor the funds available. The Funds Available infolet can be configured at a budget account level within a budget. The following infolet can be used by Chris to quickly check just the office expense funds available and drill down to research further details.

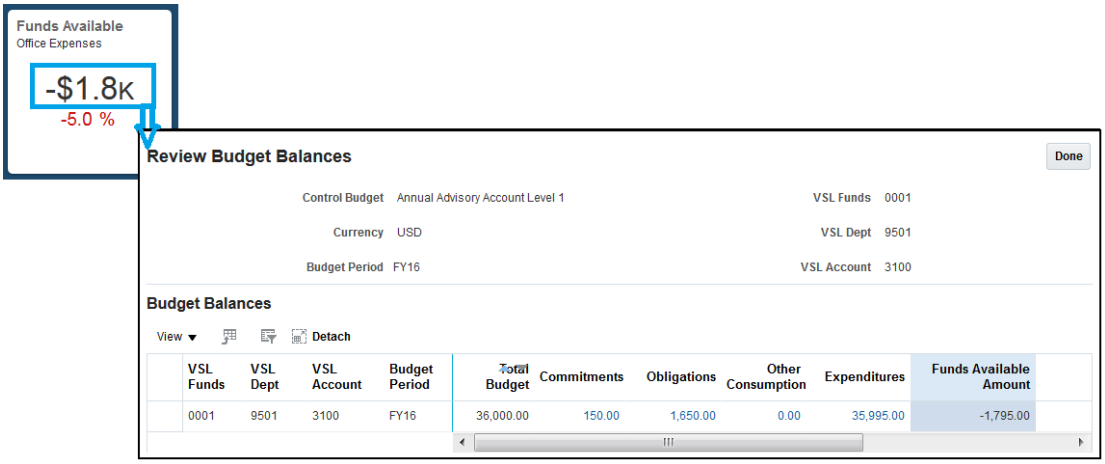


Figure 23. Chris' Home Page - Funds Available Infolet

Chris drills down to see more details of the consumption and investigate the transactions that make up the consumption amount. He can also go directly to the Budget Monitor to view the office expenses. He sees that invoice 051716-1 consumed $27,000 and drills down to review the invoice.

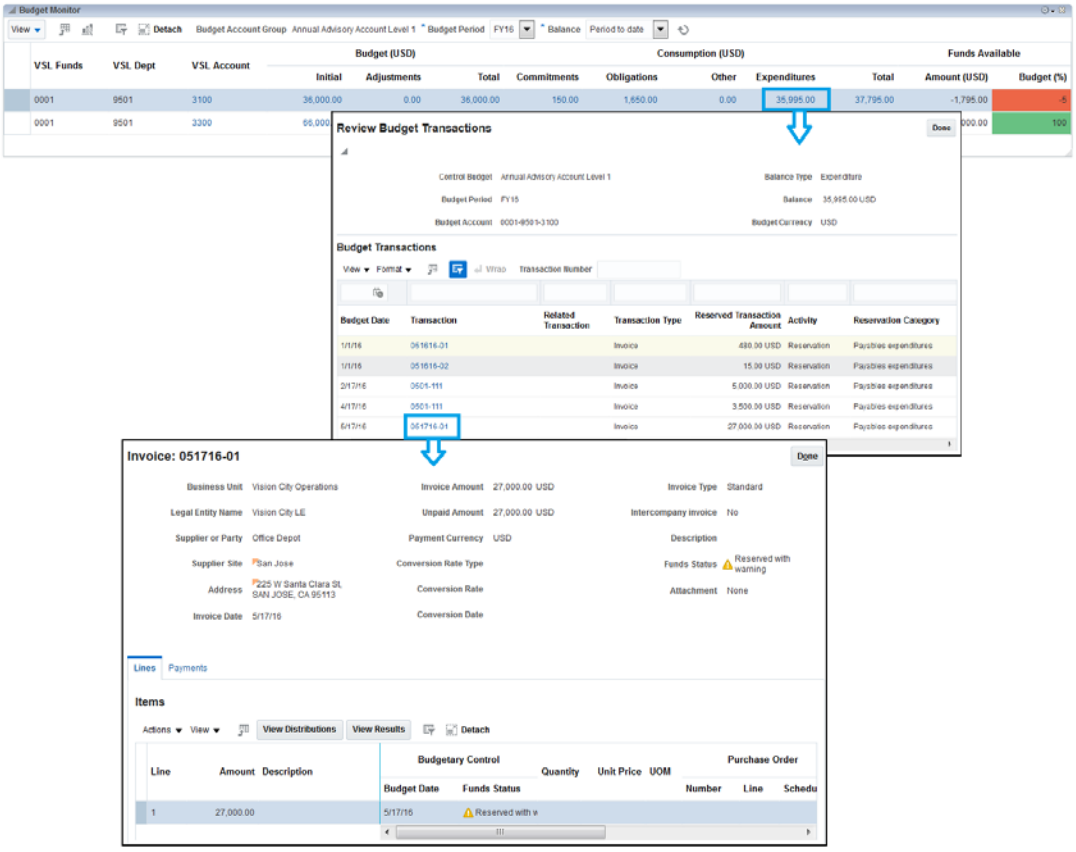


Figure 24. Budget Monitor and Drilldown

Terry or Chris can use the Control Budget Analysis Report once a month to review the funds available details for their budgets to audit for compliance and ensure there is no unexpected activity.

Control Budget	Monthly Track	Include Additional Attributes	No
Budget Period From	Jan-16	Budget Period To	Dec-16
Fund Segment From	0001	Fund Segment To	0001
Cost Center Segment From	All	Cost Center Segment To	All
Object Segment From	All	Object Segment To	All

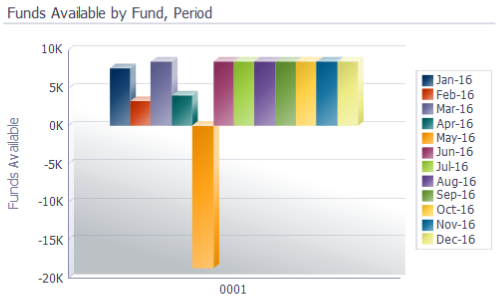
Apply

Funds Available Trend | View Funds Available | Account Activities | Funds Available Trend -... | View Funds Available - ... | Account Activities - Pr...

Ledger Vision City **Control Budget** Monthly Track **Currency** USD

Balance by Fund

	Budget Balance	Total Consumption Balance	Funds Available Percent
0001	102,000.00	37,795.00	62.95%
Total	102,000.00	37,795.00	62.95%



Balance by Cost Center and Object

			Initial Budget Balance	Budget Adjustment Balance	Commitment Balance	Obligation Balance	Other Encumbered Balance	Expenditure Balance	Total Consumption Balance	Funds Available Percent
⊟ 0001			102,000.00	0.00	150.00	1,650.00	0.00	35,995.00	37,795.00	62.95%
	⊟ 9501		102,000.00	0.00	150.00	1,650.00	0.00	35,995.00	37,795.00	62.95%
		3110	12,000.00	0.00	150.00	1,650.00	0.00	35,995.00	37,795.00	-214.96%
		3120	24,000.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00%
		3310	30,000.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00%
		3330	36,000.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00%

Figure 25. Budgetary Control Analysis Report

Encumbrance Accounting

The general accountant for the organization creates the encumbrance accounting. She creates accounting on a regular basis since encumbrance accounting is enabled for requisitions and purchase orders.

The Create Accounting program creates the following encumbrance accounting for requisition 118 for $150. The reserve for encumbrance account is defined on the Manage Budgetary Control and Encumbrance Accounting page after the ledger is enabled for encumbrance accounting.

Transaction Number 181
Event Class Requisitions
Event Type Requisition Approved

Ledger Vision City
Ledger Currency USD
Accounting Date 1/1/16
Balance Type Encumbrance

Journal Entry Description

Line	Accounting Class	Account	Encumbrance Type	Entered Currency	Entered Debit	Entered Credit	Accounted Debit	Accounted Credit
1	Requisition	0001-9501-0000-3110-000000-0000	Commitment	USD	150.00	0.00	150.00	0.00
2	Reserve for encumbrance	0001-0000-0000-7310-000000-0000	Commitment	USD	0.00	150.00	0.00	150.00
						Total for Journal Entry	**150.00**	**150.00**

Figure 26. Encumbrance Accounting for a Requisition

Purchase order 128 is created from requisition 118 for $150. The example below shows the encumbrance accounting created for the purchase order. Since it was created from a requisition, the commitment encumbrance is reversed and an obligation encumbrance is created.

Journal Entries

Transaction Number 128
Event Class Purchase Orders
Event Type Purchase Order Approved

Ledger Vision City
Ledger Currency USD
Accounting Date 1/1/16
Balance Type Encumbrance

Journal Entry Description

Line	Accounting Class	Account	Encumbrance Type	Entered Currency	Entered Debit	Entered Credit	Accounted Debit	Accounted Credit
1	Requisition	0001-9501-0000-3110-000000-0000	Commitment	USD	0.00	150.00	0.00	150.00
2	Purchase order	0001-9501-0000-3110-000000-0000	Obligation	USD	150.00	0.00	150.00	0.00
3	Reserve for encumbrance	0001-0000-0000-7310-000000-0000	Obligation	USD	0.00	150.00	0.00	150.00
4	Reserve for encumbrance	0001-0000-0000-7310-000000-0000	Commitment	USD	150.00	0.00	150.00	0.00
						Total for Journal Entry	**300.00**	**300.00**

Figure 27. Encumbrance Accounting for Purchase Order

Purchase order 128 is matched to an invoice. When accounting is created for the validated invoice, the obligation encumbrance is reversed. A journal entry for the invoice's actual expense is also recorded, which is the same accounting that would be created from the standard accrual accounting method.

Transaction Number Inv-128
Event Class Invoices
Event Type Invoice Validated

Ledger Vision City
Ledger Currency USD
Accounting Date 1/1/16
Balance Type Actual

Journal Entry Description

Line	Accounting Class	Account	Entered Currency	Entered Debit	Entered Credit	Accounted Debit	Accounted Credit
1	Liability	0001-0000-0000-7121-000000-1100	USD	0.00	150.00	0.00	150.00
2	Item expense	0001-9501-0000-3110-000000-0000	USD	150.00	0.00	150.00	0.00
					Total for Journal Entry	**150.00**	**150.00**

Transaction Number Inv-128
Event Class Invoices
Event Type Invoice Validated

Ledger Vision City
Ledger Currency USD
Accounting Date 1/1/16
Balance Type Encumbrance

Journal Entry Description

Line	Accounting Class	Account	Encumbrance Type	Entered Currency	Entered Debit	Entered Credit	Accounted Debit	Accounted Credit
1	Purchase order	0001-9501-0000-3110-000000-0000	Obligation	USD	0.00	150.00	0.00	150.00
2	Reserve for encumbrance	0001-0000-0000-7310-000000-0000	Obligation	USD	150.00	0.00	150.00	0.00
						Total for Journal Entry	**150.00**	**150.00**

Figure 28. Invoice Accounting

Other Considerations

Manage Budgetary Control Choices

Certain options have critical implications on determining how Oracle ERP Cloud Budgetary Control handles your transactions. Consider the following options when enabling budgetary control:

- Control Budget Filter
- Fail Budgetary Control for Budget Dates Not in Valid Budget Date Range
- Default Date Rule

For more information, see the help section of the Edit Budgetary Control and Encumbrance Accounting page.

Supplemental Rules

Supplemental rules create exceptions to the control level and tolerance settings defined in the header of the Create/Edit Control Budget page. Use supplemental rules to further control how a subset of budget dimensions are validated. In the implementation example, Terry doesn't want to control spending for the Total Cost of Goods Sold (1800) and Personnel (2000) expenses in the Annual Advisory Account Level 1 budget. A supplemental rule can be created with a control level of **None** so these expenses (1800 and 2000) won't be consumed against the budget.

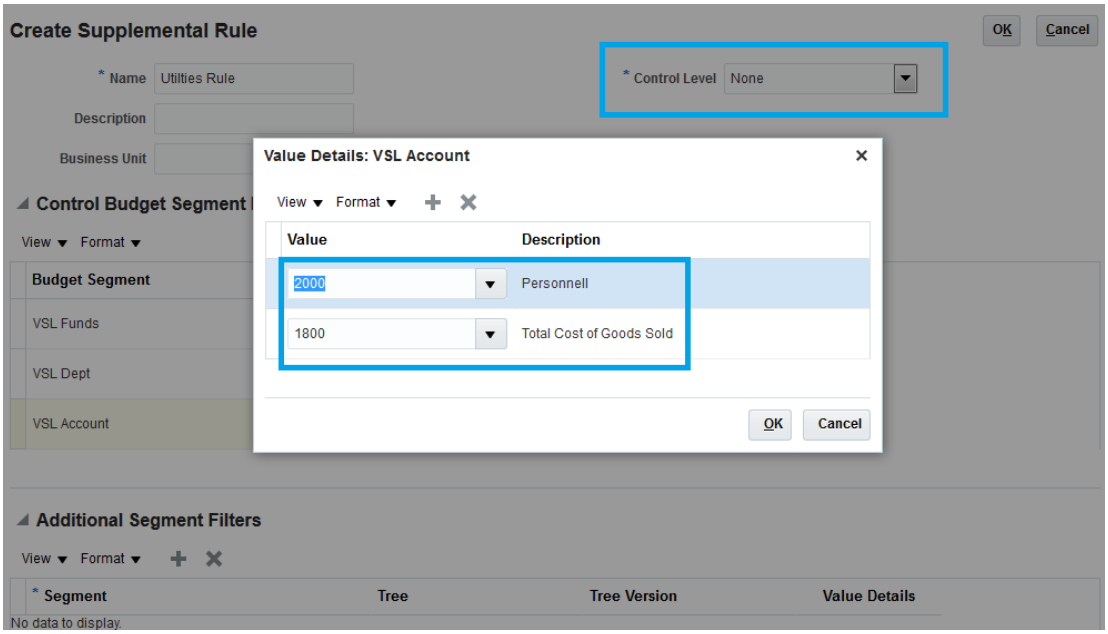


Figure 29. Supplemental Rule

For more information on supplemental rules, see the help section on the Create Control Budget page.

Source Detail and Summary Control Budgets

Two control budgets can be linked and have a source detail/summary relationship if the control budget structure and start/end dates of the summary control budget are equal to or a subset of the source detail control budget. Once you enter the budget data into the source detail control budget, the summary control budget is automatically updated.

The implementation example and other examples in this document use this linked relationship. The Monthly Track control budget is the source detail budget, and the two annual control budgets use the summarized budget amounts of the Monthly Track control budget. The control budgets are tied together by naming the detail control budget when defining the summary control budget. In the following example, the Monthly Track control budget is specified as the source budget name of the Annual Advisory Account Level 1 control budget.

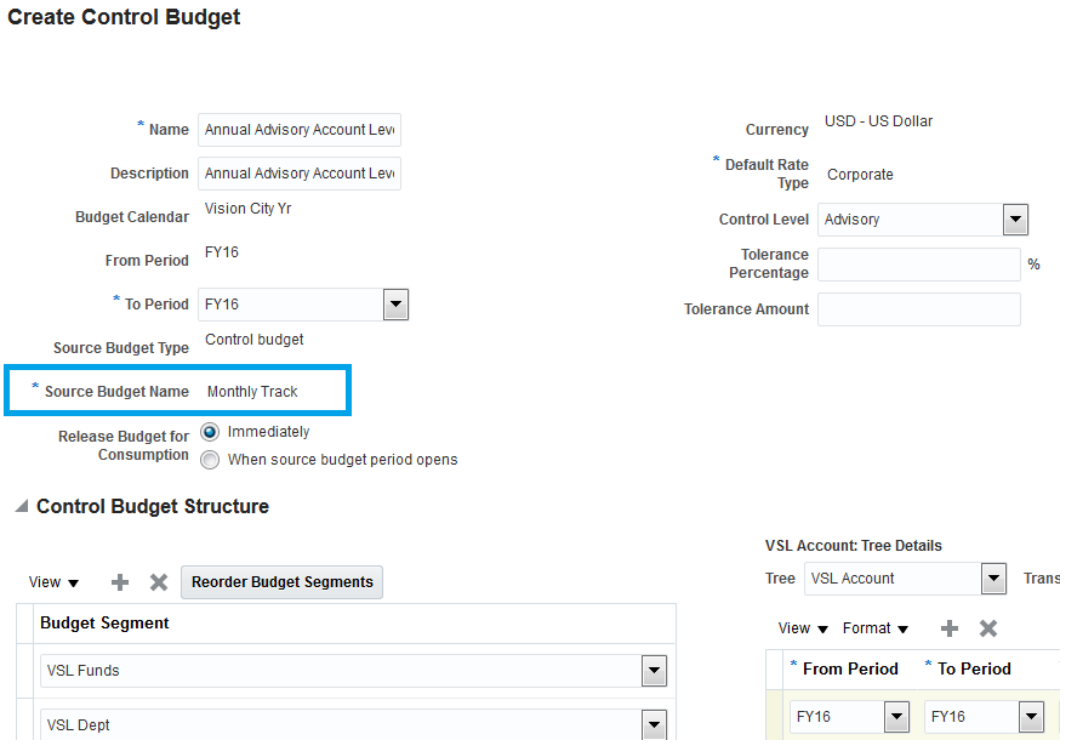


Figure 30. Source Budget Name

A source detail control budget can have multiple summary control budgets. However, a summary control budget can't be a source for another control budget. The source detail control budget can accept budget data from any source (such as Hyperion Planning, spreadsheet, and third party), and all budget data entered into the source detail control budget is propagated to the summary control budget. Summary control budgets can't accept data from any source other than the source detail control budget.

The summary control budget can control using different segments as long as they are a subset of the segments in the source detail control budget. The summary control budget can control using a different hierarchy level than the source detail control budget, provided that the source detail control budget is more detailed.

Release Budget for Consumption

When using source detail and summary control budgets, you can prevent budget consumption in future periods by controlling the release of budget amounts by period. **Release Budget for Consumption** is available on control budgets that have a source budget type of **Control budget**. Depending on how you set the **Release Budget for Consumption** option, you can release budget amounts:

- When the period of the specified source control budget is opened
- Upon import to the source control budget

In the implementation example, the Annual Advisory Account Level 1 control budget set the **Release Budget for Consumption** option to release budget amounts immediately when budget amounts are loaded into the Monthly Track control budget. Therefore the $12,000 that was imported into the Monthly Track control budget for budget account 0001-9501-3110 was immediately available in the Annual Advisory Account Level 1 control budget.

If you use the other option – release for consumption when the Monthly Track budget period opens, you can decide when you want to release funds into the Annual Advisory Account Level 1 control.

The funds available balance impact on the Annual Advisory Account Level 1 control budget based on this option is illustrated in the table below.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Budget Period of Monthly Track	Available for Budgeting											
Imported Budget Amounts into Monthly Track	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
Balance for 0001-9501-3110 in Annual Advisory Account Level 1 for FY16 Upon Budget Import if Release Budget for Consumption is:												
Immediately	12,000											
When Source Budget Period Open	0											
Balance for 0001-9501-3110 in Annual Advisory Account Level 1 for FY16 When Jan-16 budget period is open if Release Budget for Consumption is:												
Immediately	12,000											
When Source Budget Period Open	1,000											

Figure 31. Release Budget for Consumption Option

Reporting Trees

Starting in R11, you can add tree hierarchies to an Essbase cube to be used solely for reporting. In our implementation example, the monthly control budgets didn't share the same Essbase cube as the yearly control budgets since they use different budget calendars. Budgetary Control maintains separate cubes for every unique combination of budget calendar and control budget structure. The Monthly Track control budget is set up at a detail level so that the summary of budget amounts isn't possible. By running the Add Reporting Tree to Budgetary Control Cubes process, you can add a tree hierarchy, such as the VSL Account Tree, to an Essbase cube to enable summary reporting for the budget.

Implementing During a Fiscal Year

When implementing Budgetary Control after the budget year has begun, consider the impact of existing General Ledger balances. Since some of these balances would have consumed your budget and are therefore not reflected in budgetary control, adjust the budget amount so that the funds available balance is accurate.

Conclusion

Carefully consider your requirements when you implement Budgetary Control and encumbrance accounting. By implementing a flexible, integrated, and complete configuration, you can meet your budget, budgetary control, accounting and reporting needs.

Additional Resources

For more information on implementing Budgetary Control and encumbrance accounting, see the Oracle ERP Cloud Budgetary Control and Encumbrance Accounting Implementation Guide, Doc ID 2022150.1, on My Oracle Support.

For new features and release enhancements, see the following:

- Enterprise Resource Planning Release Readiness page
- New Features in Oracle Budgetary Control presentation on Oracle Applications Customer Connect

Oracle Corporation, World Headquarters
500 Oracle Parkway
Redwood Shores, CA 94065, USA

Worldwide Inquiries
Phone: +1.650.506.7000
Fax: +1.650.506.7200

CONNECT WITH US

blogs.oracle.com/oracle

facebook.com/oracle

twitter.com/oracle

oracle.com

Integrated Cloud Applications & Platform Services

Copyright © 2016, Oracle and/or its affiliates. All rights reserved. This document is provided *for* information purposes only, and the contents hereof are subject to change without notice. This document is not warranted to be error-free, nor subject to any other warranties or conditions, whether expressed orally or implied in law, including implied warranties and conditions of merchantability or fitness for a particular purpose. We specifically disclaim any liability with respect to this document, and no contractual obligations are formed either directly or indirectly by this document. This document may not be reproduced or transmitted in any form or by any means, electronic or mechanical, for any purpose, without our prior written permission.

Oracle and Java are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

Intel and Intel Xeon are trademarks or registered trademarks of Intel Corporation. All SPARC trademarks are used under license and are trademarks or registered trademarks of SPARC International, Inc. AMD, Opteron, the AMD logo, and the AMD Opteron logo are trademarks or registered trademarks of Advanced Micro Devices. UNIX is a registered trademark of The Open Group. 0116

Budgetary Control and Encumbrance Accounting Configuration Best Practices
June 2016
Author: Cynthia Bilbie
Contributing Authors: Anne Wong

Oracle is committed to developing practices and products that help protect the environment